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Notifications

3. Fibonacci sequence

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$F_0 = 1, \quad F_1 = 1, \quad \text{and} \quad F_{n+1} = F_n + F_{n-1} \quad \text{for } n \geq 1.$

In this problem, we'll use eigenvalues to compute an explicit formula for and describe the growth rate of F_n .

Fibonacci part (a)

1.0/1 point (graded)
Find a 2×2 matrix $\mathbf{M}$ such that, for any $k \geq 1$,

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \mathbf{M} \begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix}.$$

(Enter the matrix as a list of lists. That is enter each row between square brackets, and separate rows by commas. Surround the entire matrix by square brackets; e.g. type `[[a, b],[c, d]]` for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Note when typed correctly it will render as a matrix beneath the answer box!)

$\mathbf{M} =$

✓ Answer: [[1, 1],[1, 0]]

Solution:

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \begin{pmatrix} F_k + F_{k-1} \\ F_k \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix},$$

so $\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$

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❗ Answers are displayed within the problem

We can compute F_n using a simple matrix expression:

$$= \mathbf{M} \begin{pmatrix} F_{n-1} \\ F_{n-2} \end{pmatrix} = \mathbf{M}^2 \begin{pmatrix} F_{n-2} \\ F_{n-3} \end{pmatrix} = \dots = \mathbf{M}^{n-2} \begin{pmatrix} F_2 \\ F_1 \end{pmatrix} = \mathbf{M}^{n-1} \begin{pmatrix} F_1 \\ F_0 \end{pmatrix} = \mathbf{M}^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The goal of the rest of the problem will be to compute $\mathbf{M}^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and thus F_n .

Part (b)

2/2 points (graded)
Find the eigenvalues $\lambda_1 > \lambda_2$. (Enter the larger eigenvalue in the first answer box. Give an analytic expression rather than a numerical one.)

$\lambda_1 =$

✓ Answer: (1+sqrt(5))/2

$$\frac{1+\sqrt{5}}{2}$$

$\lambda_2 =$

✓ Answer: (1-sqrt(5))/2

$$\frac{1-\sqrt{5}}{2}$$

Solution:

The eigenvalues are the roots of the characteristic polynomial

$$\det(\lambda \mathbf{I} - \mathbf{M}) = \det\left(\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\right) = \det\begin{pmatrix} \lambda - 1 & -1 \\ -1 & \lambda \end{pmatrix}$$
$$= (\lambda - 1)\lambda - (-1)(-1) = \lambda^2 - \lambda - 1.$$

Its roots are given by

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{1 - \sqrt{5}}{2}.$$

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Part (c)

2/2 points (graded)
Find the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of $\mathbf{M}$ corresponding to eigenvalues $\lambda_1 > \lambda_2$. In particular, find the numbers u and w so that

$$\mathbf{v}_1 = \begin{pmatrix} u \\ 1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} w \\ 1 \end{pmatrix}.$$

Hint: Use the fact that $\lambda^2 - \lambda - 1 = 0$ can be rearranged to show $\lambda - 1 = 1/\lambda$. Use this in your computations to simplify algebra.

(Note that $\mathbf{v}_1$ is the eigenvector corresponding to the larger eigenvalue.)

$u =$

Answer: (1+sqrt(5))/2

$w =$

Answer: (1-sqrt(5))/2

Solution:

With λ equal to either λ_1 or λ_2 we have

$$\lambda - 1 = 1/\lambda.$$

The eigenvectors are the elements in the nullspace of the matrix $\lambda \mathbf{I} - \mathbf{M}$.

$$\text{NS}(\lambda \mathbf{I} - \mathbf{M}) = \text{NS}\begin{pmatrix} \lambda - 1 & -1 \\ -1 & \lambda \end{pmatrix} = \text{NS}\begin{pmatrix} 1/\lambda & -1 \\ -1 & \lambda \end{pmatrix}.$$

Subtracting a multiple of row 1 from row 2 does not change the nullspace, therefore

$$\text{NS}\begin{pmatrix} 1/\lambda & -1 \\ -1 & \lambda \end{pmatrix} = \text{NS}\begin{pmatrix} 1/\lambda & -1 \\ 0 & 0 \end{pmatrix}.$$

Therefore $\begin{pmatrix} \lambda \\ 1 \end{pmatrix}$ is an eigenvector.

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Part (d)

2/2 points (graded)

Write the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ as a linear combination $a\mathbf{v}_1 + b\mathbf{v}_2$ of the eigenvectors.

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$a =$ ✓ Answer: (1+sqrt(5))/(2*sqrt(5))

$\frac{\sqrt{5}+1}{2\cdot\sqrt{5}}$

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we want

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = a \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix} + b \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix},$$

so a and b satisfy the system of linear equations

$$\begin{aligned} a\lambda_1 + b\lambda_2 &= 1 \\ a + b &= 1. \end{aligned}$$

Solving this gives

$$a = \frac{1 - \lambda_2}{\lambda_1 - \lambda_2} = \frac{1 + \sqrt{5}}{2\sqrt{5}} \quad \text{and} \quad b = 1 - a = \frac{-1 + \sqrt{5}}{2\sqrt{5}}.$$

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Answers are displayed within the problem

Part (e)

1.0/1 point (graded)
Use your answers from parts (a)-(d) to find a formula for F_n . The formula should be something you can compute directly, not something that writes F_n in terms of other F_j .

$F_n =$ ✓

Answer: (1/sqrt(5))* ((1+sqrt(5))/2)^(n+1) - ((1-sqrt(5))/2)^(n+1))

Solution:

We have

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = a\mathbf{v}_1 + b\mathbf{v}_2,$$

and

$$\begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} = \mathbf{M}^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Expanding this out and using the fact that $\mathbf{M}^k \mathbf{v}_i = \lambda_i^k \mathbf{v}_i$, we get

$$\begin{aligned} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} &= \mathbf{M}^{n-1} (a\mathbf{v}_1 + b\mathbf{v}_2) \\ &= a\lambda_1^{n-1}\mathbf{v}_1 + b\lambda_2^{n-1}\mathbf{v}_2 \\ &= a\lambda_1^{n-1} \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix} + b\lambda_2^{n-1} \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix} \\ &= \frac{1+\sqrt{5}}{2} \begin{pmatrix} \lambda_1^n \\ \lambda_1^{n-1} \end{pmatrix} + \frac{-1+\sqrt{5}}{2} \begin{pmatrix} \lambda_2^n \\ \lambda_2^{n-1} \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
 &= \frac{\lambda_1}{\sqrt{5}} \begin{pmatrix} \lambda_1^n \\ \lambda_1^{n-1} \end{pmatrix} - \frac{\lambda_2}{\sqrt{5}} \begin{pmatrix} \lambda_2^n \\ \lambda_2^{n-1} \end{pmatrix} \\
 &= \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_1^{n+1} - \lambda_2^{n+1} \\ \lambda_1^n - \lambda_2^n \end{pmatrix}.
 \end{aligned}$$

Taking the first coordinate gives our formula for F_n :

$$F_n = \frac{1}{\sqrt{5}}(\lambda_1^{n+1} - \lambda_2^{n+1}) = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{n+1} \right].$$

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You have used 4 of 15 attempts

i Answers are displayed within the problem

Part (f)

1/1 point (graded)
Find $\ln F_{1,000,000}$ to 10 decimal places.

(Use [MATLAB online](#). Note that in MATLAB, typing **log** gives the natural logarithm, not the base 10 logarithm. Observe that MATLAB typically only displays four decimal places, but stores a larger number. To see the larger number of decimal digits type **format long** before you display numbers of interest.)

481211.5015524723

✔ Answer: 481211.5015524723

Solution:

In the previous problem you found that

$$F_n = \frac{1}{\sqrt{5}}(\lambda_1^{n+1} - \lambda_2^{n+1}).$$

The term $-\lambda_2^{n+1}$ is negligible for large n since $|\lambda_2| < 1$; therefore

$$\ln F_n \approx \ln \frac{1}{\sqrt{5}} + (n+1) \ln(\lambda_1).$$

We compute this directly using MATLAB for $n = 10^6$.

Note that if you try to compute $\ln F_{10^6}$ directly, MATLAB will output `inf`. Instead, you must first observe that only the term involving the larger eigenvalue will contribute most. If you omit the $\ln(1/\sqrt{5})$ term your answer will not be correct up to the desired number of decimal digits.

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You have used 1 of 10 attempts

i Answers are displayed within the problem

Part (g)

2/2 points (graded)
Consider a generalized fibonacci G sequence $1, 1, 1, 3, 5, 9, 17, \dots$ is created by summing the last 3 entries in the sequence together:

$$\begin{aligned}
 G_0 &= 1, \quad G_1 = 1, \quad G_2 = 1, \quad \text{and} \\
 G_{n+1} &= G_n + G_{n-1} + G_{n-2} \text{ for } n \geq 2.
 \end{aligned}$$

Find a 3×3 matrix $\mathbf{M}$ such that, for any $k \geq 2$,

$$\begin{pmatrix} G_{k+1} \\ G_k \\ G_{k-1} \end{pmatrix} = \mathbf{M} \begin{pmatrix} G_k \\ G_{k-1} \\ G_{k-2} \end{pmatrix}.$$

Use [MATLAB online](#) to find a numerical value for G_{25} .

$G_{25} =$

1800281

✔ Answer: 1800281

Find $\lim_{n \rightarrow \infty} \frac{\ln G_n}{n}$ to 10 decimal places.

$\lim_{n \rightarrow \infty} \frac{\ln G_n}{n} =$

0.6093778634

✔ Answer: 0.609377863436006

Solution:

In this case $\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$. Since

$$\begin{pmatrix} G_3 \\ G_2 \\ G_1 \end{pmatrix} = \mathbf{M} \begin{pmatrix} G_2 \\ G_1 \\ G_0 \end{pmatrix},$$

$$\begin{pmatrix} G_4 \\ G_3 \\ G_2 \end{pmatrix} = \mathbf{M}^2 \begin{pmatrix} G_2 \\ G_1 \\ G_0 \end{pmatrix},$$

it follows that G_{25} is the first entry in the vector given by $\mathbf{M}^{23} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Plugging into MATLAB for use as a calculator gives the answer directly.

To find $\lim_{n \rightarrow \infty} \frac{\ln G_n}{n}$, we use an approach analogous to what we did with the Fibonacci sequence. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be eigenvectors of $\mathbf{M}$ so that λ_1 is the largest eigenvalue.

We start by expressing the vector representing the first three numbers as a linear combination of the eigenvectors

$$\begin{pmatrix} G_2 \\ G_1 \\ G_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3.$$

We write

$$\begin{aligned} \begin{pmatrix} G_n \\ G_{n-1} \\ G_{n-2} \end{pmatrix} &= \mathbf{M}^{n-2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\ &= \mathbf{M}^{n-2} (a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3) \\ &= a_1 \lambda_1^{n-2} \mathbf{v}_1 + a_2 \lambda_2^{n-2} \mathbf{v}_2 + a_3 \lambda_3^{n-2} \mathbf{v}_3 \\ &\approx a_1 \lambda_1^{n-2} \mathbf{v}_1 \end{aligned}$$

The number G_n is the first entry of this vector, which is

$$G_n = b_1 \lambda_1^{n-2},$$

where $b_1 = a_1$ times the first entry of $\mathbf{v}_1$. Taking the natural logarithm and dividing by n we get

$$\frac{\ln G_n}{n} = \frac{\ln b_1 + (n-2) \ln(\lambda_1)}{n}.$$

As n tends to infinity, this tends to $\ln(\lambda_1)$, which we compute using MATLAB. (In this case the constant term $\ln b_1/n$ is zero in the limit.)

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3. Fibonacci sequence

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